

Introduction to Robotics

CSE 461

Class Topic: Introduction to Control System Theory (PID)

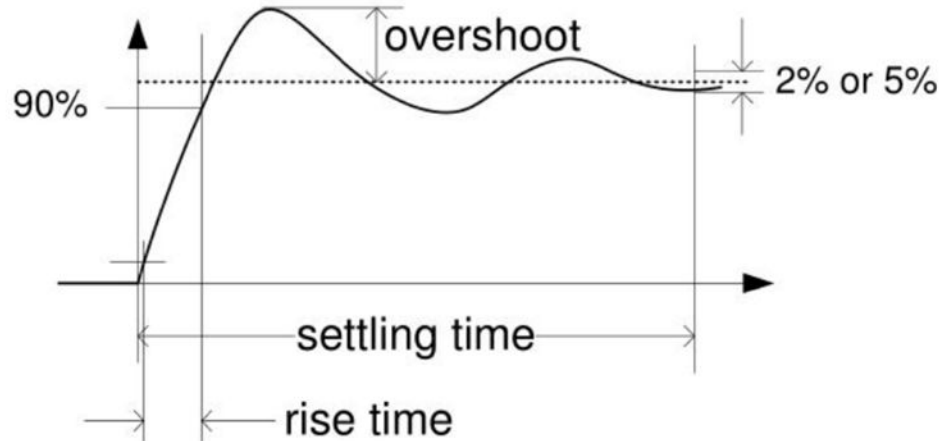
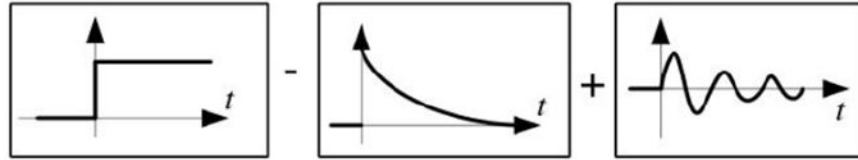
Riad Ahmed

Lecturer

Brac University

2nd order control system

- Typical response to step input is:



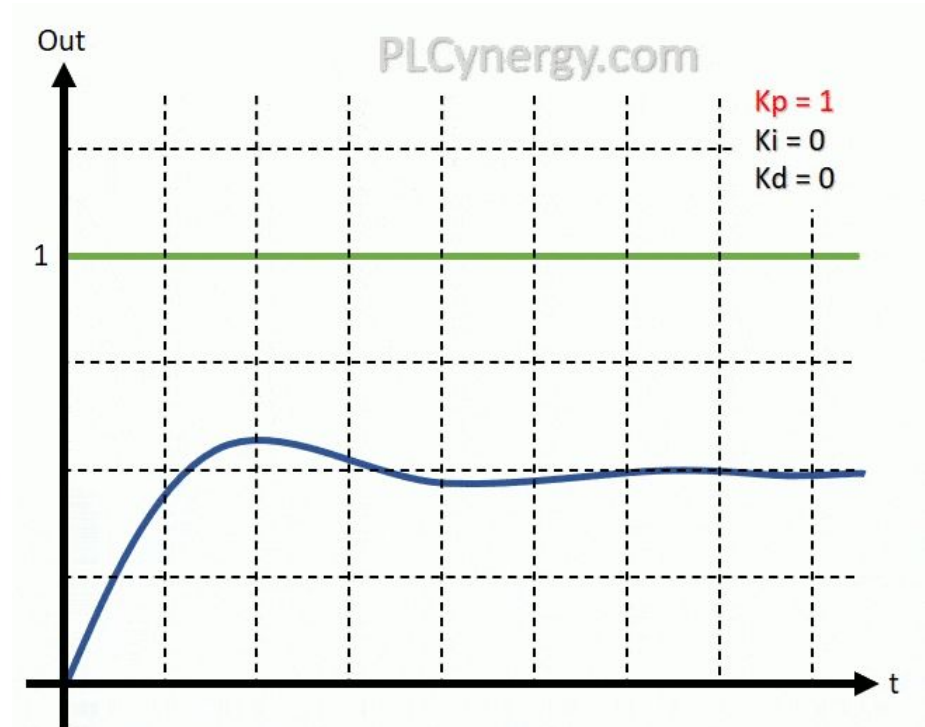
overshoot -- % of final value exceeded at first *oscillation*

rise time -- time to span from 10% to 90% of the final value

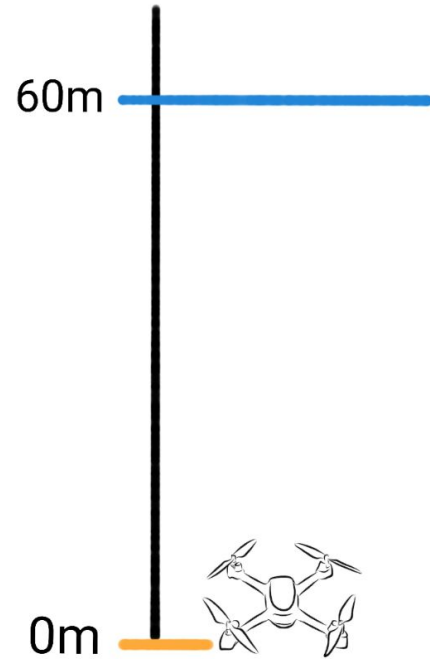
settling time -- time to reach within 2% or 5% of the final value

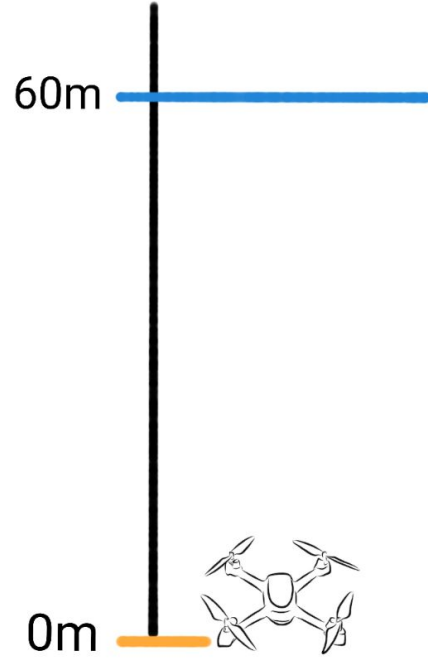
Effect of PID controller function

- Proportional Action
 - Simplest Controller Function
- Integral Action
 - Eliminates steady-state error
 - Can cause oscillations
- Derivative Action (“rate control”)
 - Effective in transient periods
 - Provides faster response (higher sensitivity)
 - Never used alone

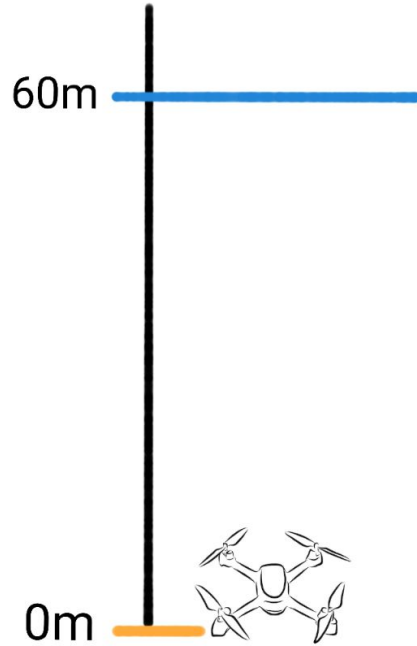


<https://www.youtube.com/watch?v=wkfEZmsQqiA>



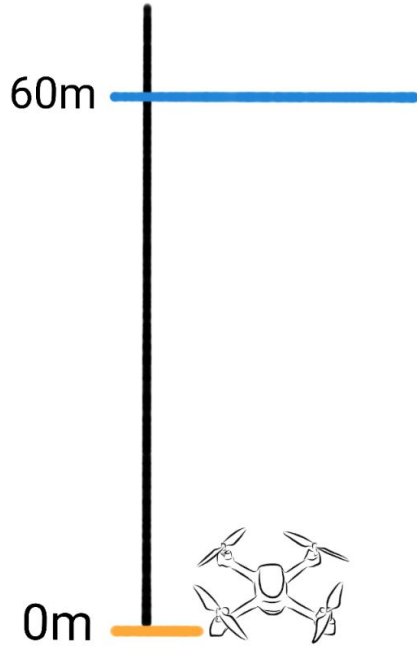


Desired Altitude = 60m
Actual Altitude = 0m



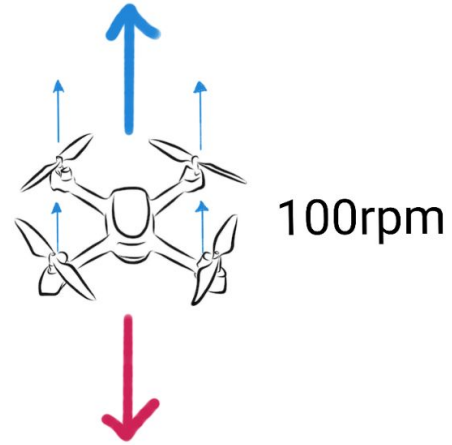
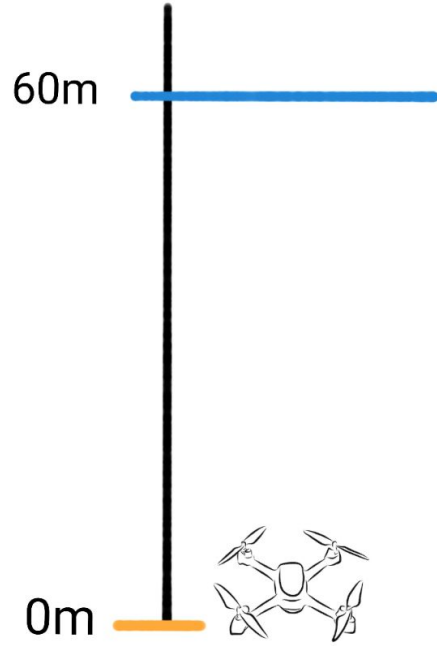
Desired Altitude = 60m
Actual Altitude = 0m

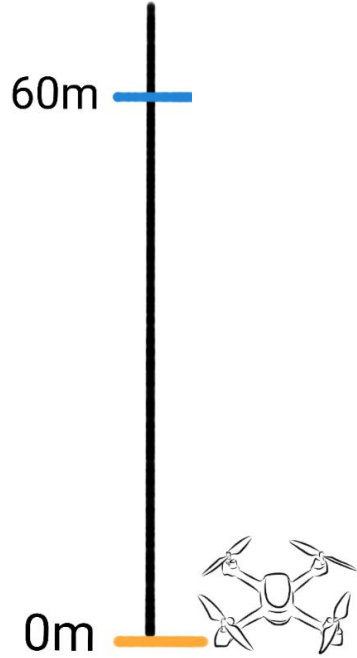
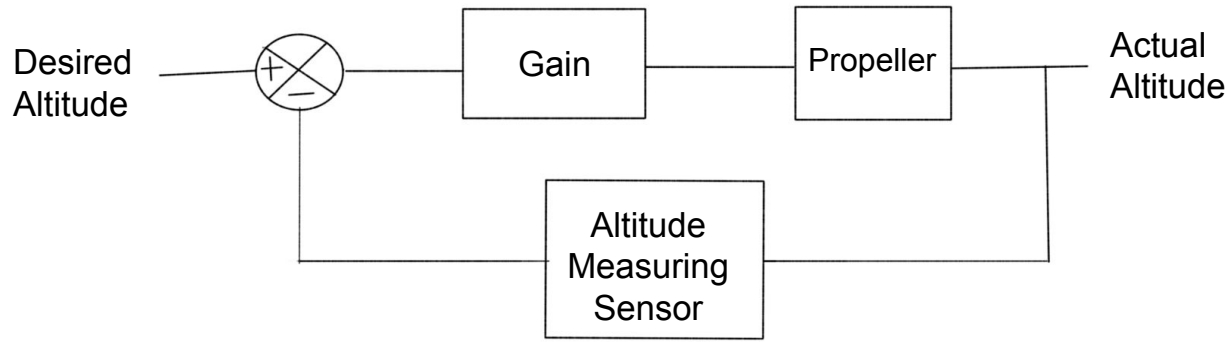
$$\begin{aligned} \text{Desired Altitude} - \text{Actual Altitude} \\ &= 60\text{m} - 0\text{m} \\ &= 60 \end{aligned}$$

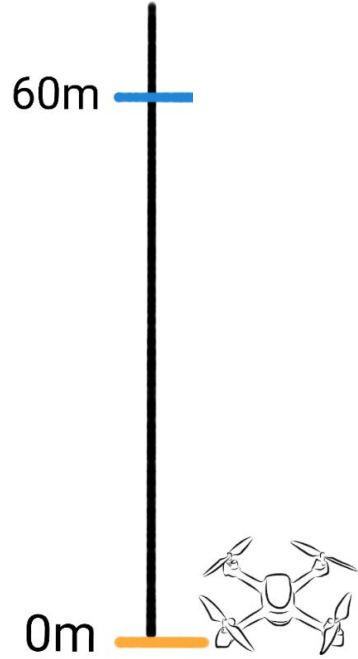
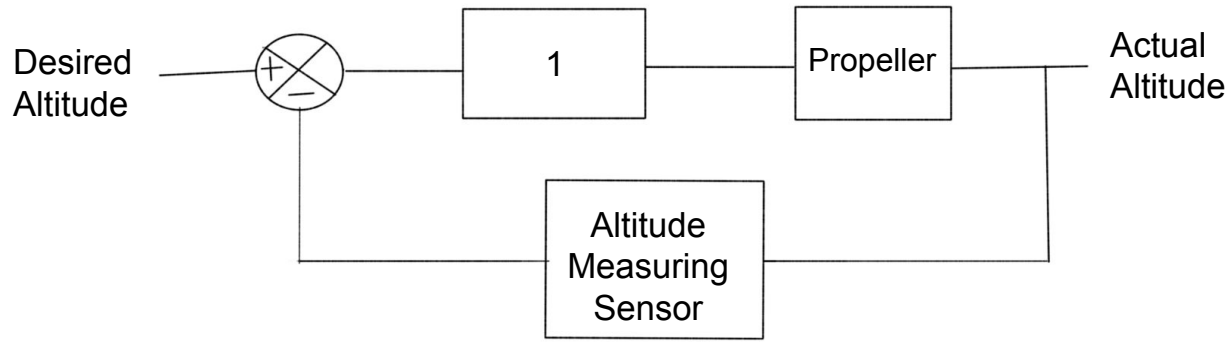


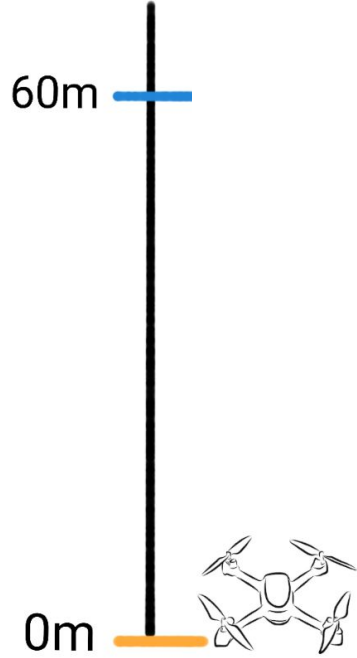
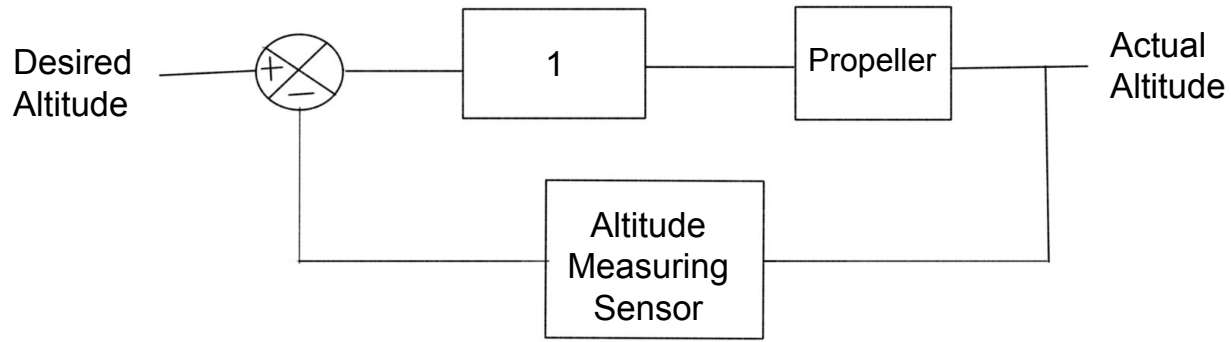
Desired Altitude = 60m
Actual Altitude = 0m

Desired Altitude - Actual Altitude
= 60m - 0m
= 60
= Error



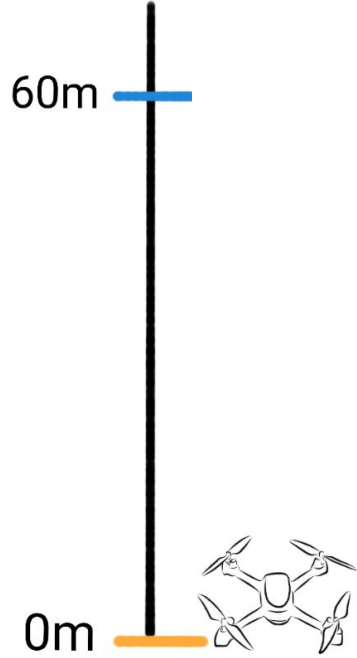
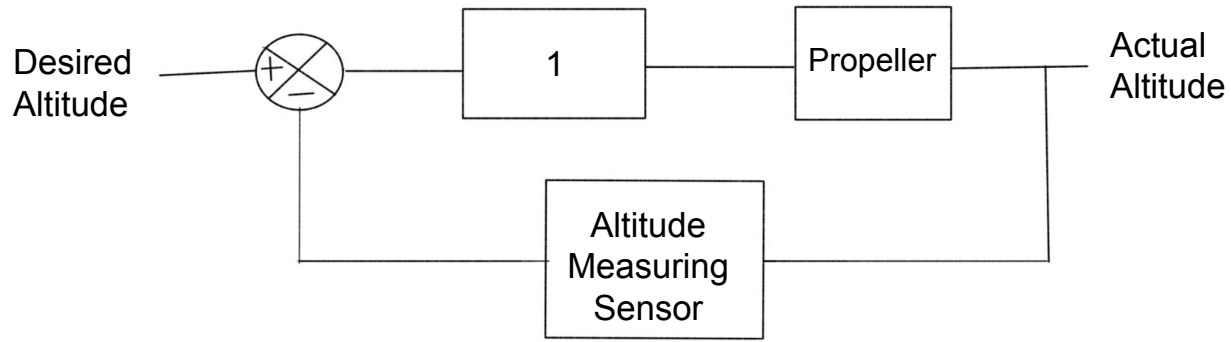






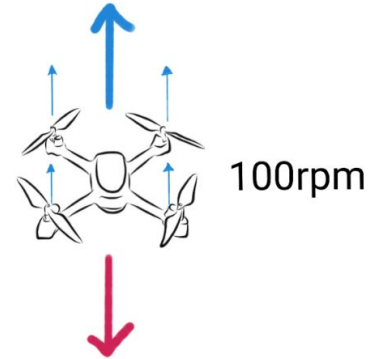
$$\begin{aligned}\text{Desired Altitude} - \text{Actual Altitude} \\ &= 60\text{m} - 0\text{m} \\ &= 60\end{aligned}$$

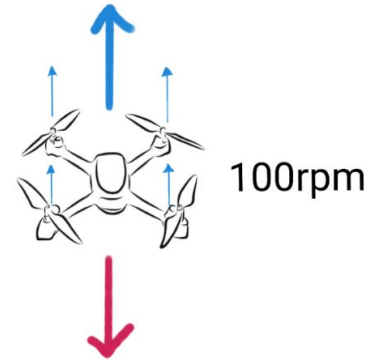
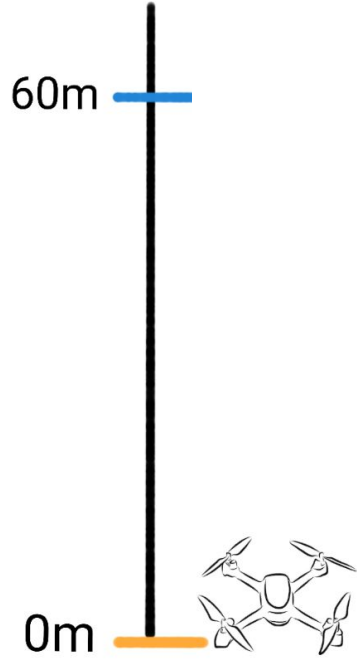
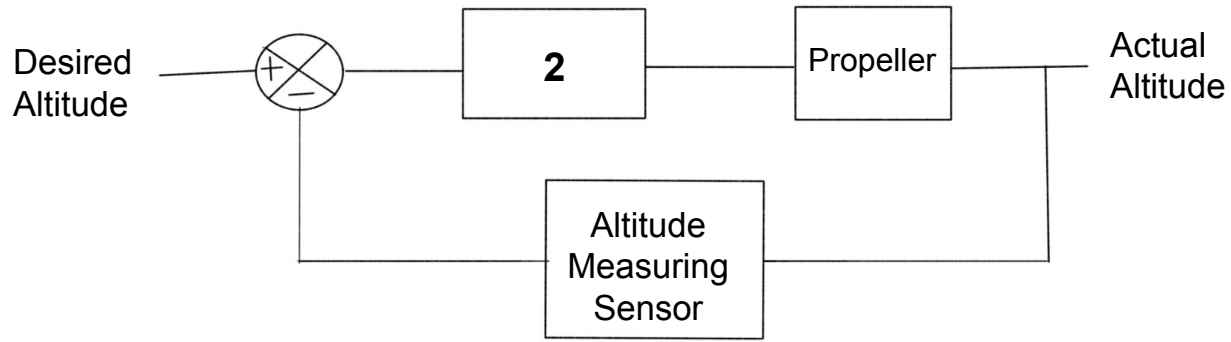
$$\text{Propeller Speed} = 60 * 1 = 60 \text{ rpm}$$

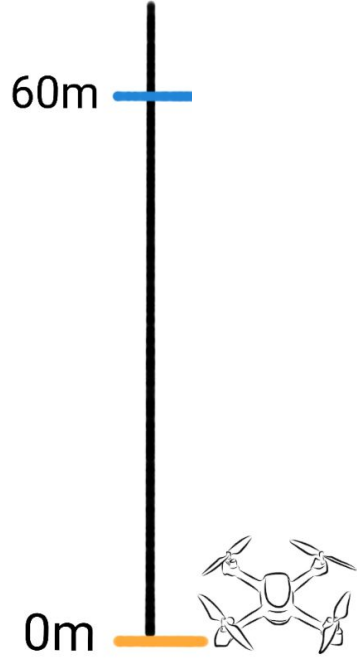
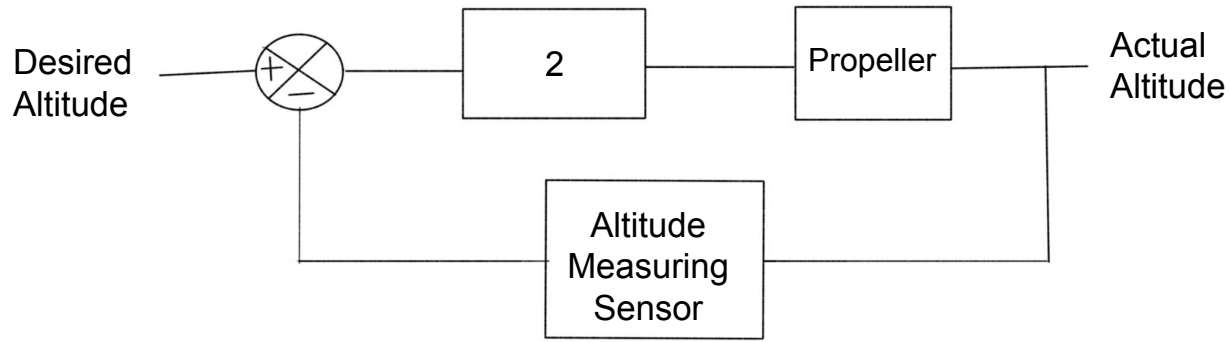


$$\begin{aligned}\text{Desired Altitude} - \text{Actual Altitude} \\ &= 60\text{m} - 0\text{m} \\ &= 60\end{aligned}$$

$$\text{Propeller Speed} = 60 * 1 = 60 \text{ rpm}$$

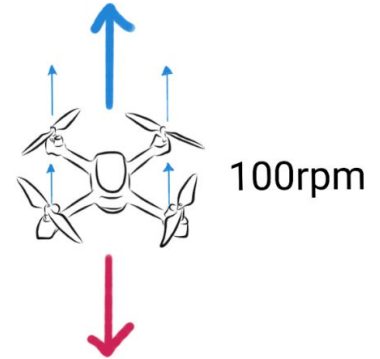


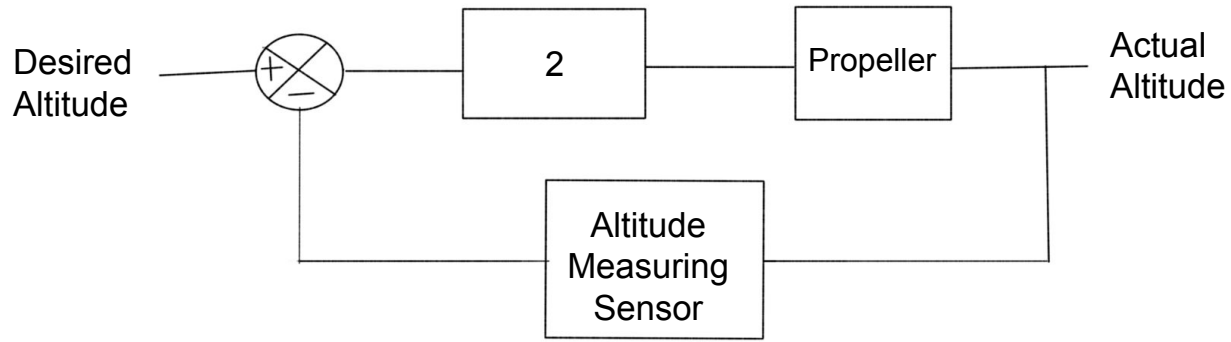




$$\begin{aligned}\text{Desired Altitude} - \text{Actual Altitude} \\ &= 60\text{m} - 0\text{m} \\ &= 60\end{aligned}$$

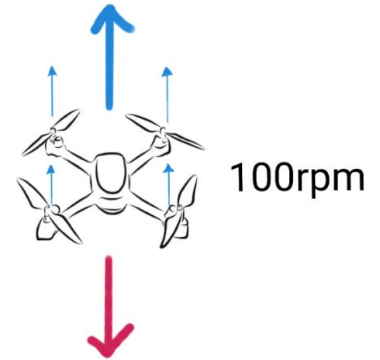
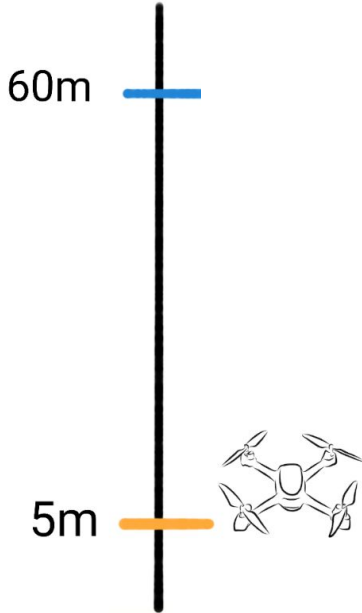
$$\text{Propeller Speed} = 60 * 2 = 120 \text{ rpm}$$

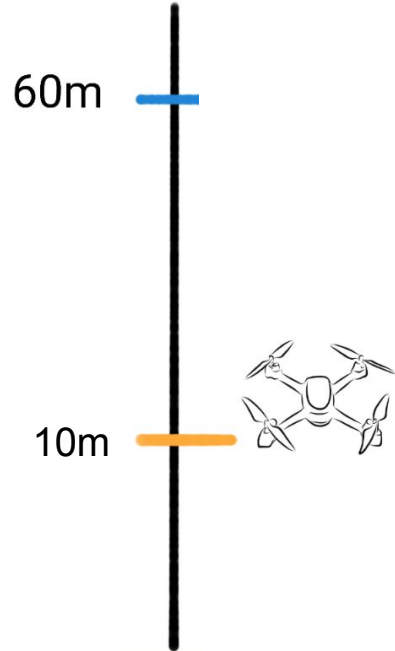
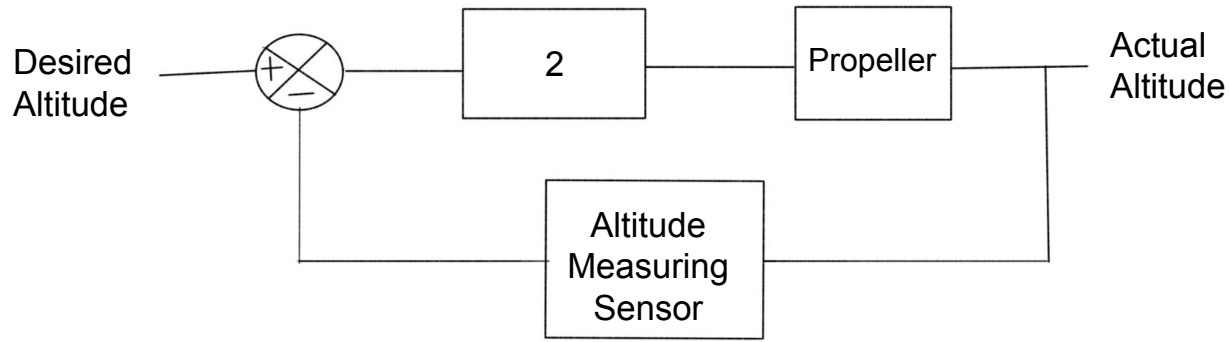




$$\begin{aligned}\text{Desired Altitude} - \text{Actual Altitude} &= 60\text{m} - 5\text{m} \\ &= 55\end{aligned}$$

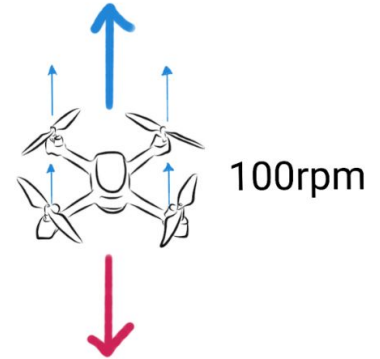
$$\text{Propeller Speed} = 55 * 2 = 110 \text{ rpm}$$

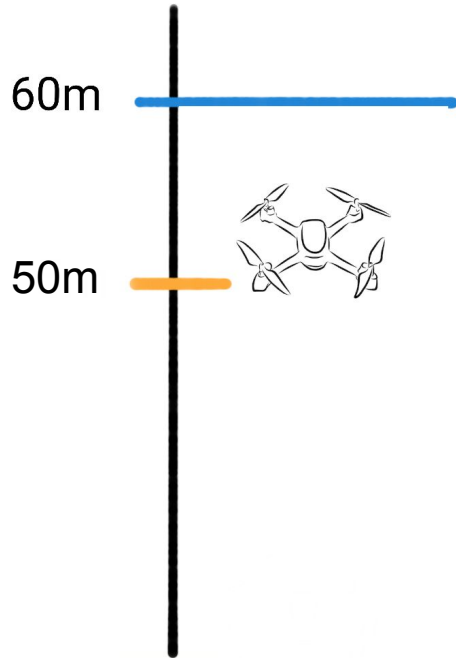
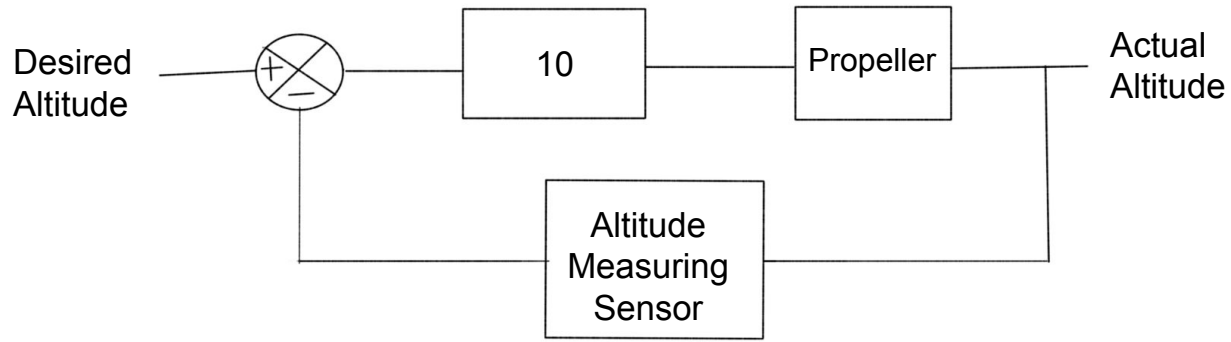




$$\begin{aligned}\text{Desired Altitude} - \text{Actual Altitude} \\ &= 60\text{m} - 10\text{m} \\ &= 50\end{aligned}$$

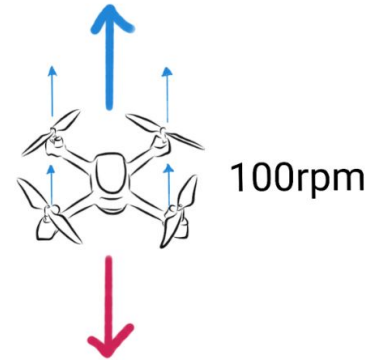
$$\text{Propeller Speed} = 50 * 2 = 100 \text{ rpm}$$

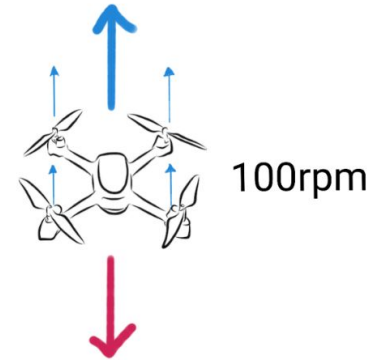
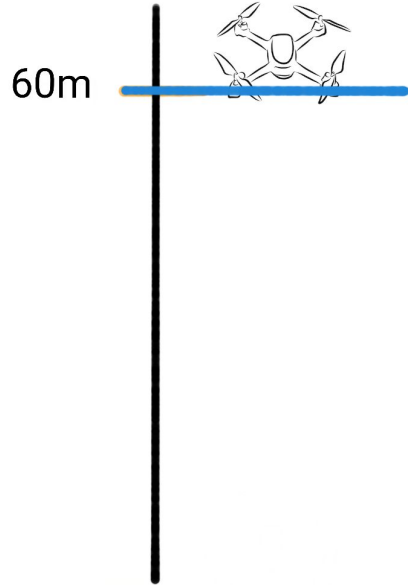
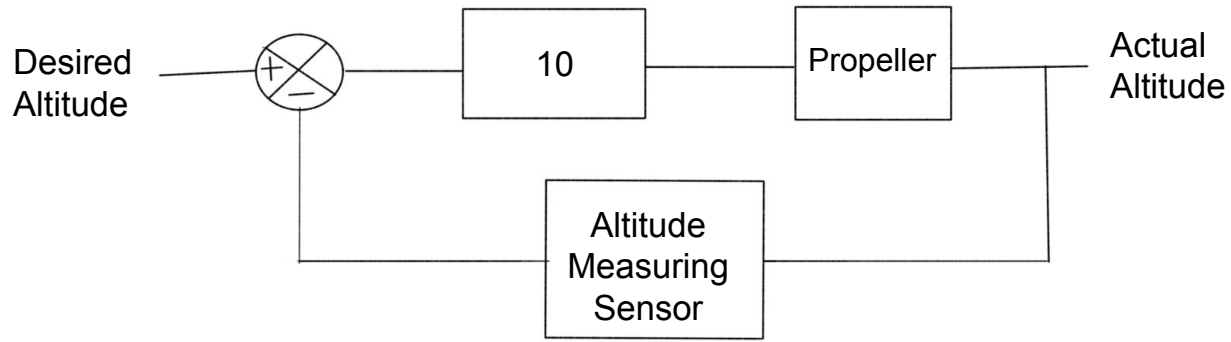


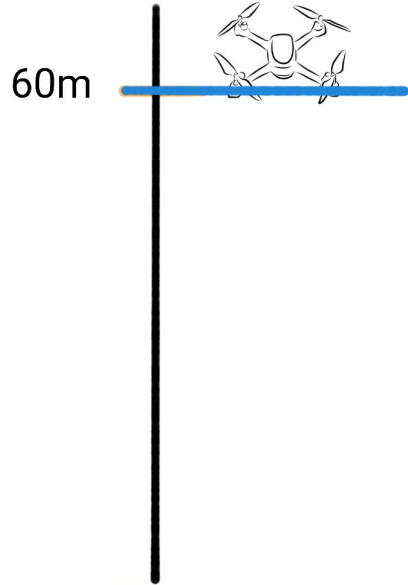
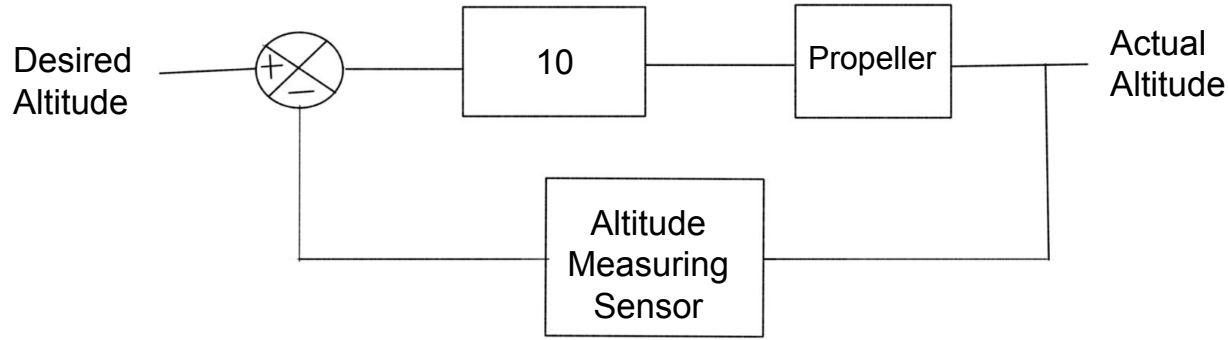


$$\text{Propeller Speed} = 10 * 10 = 100 \text{ rpm}$$

$$\text{Error} = 10$$

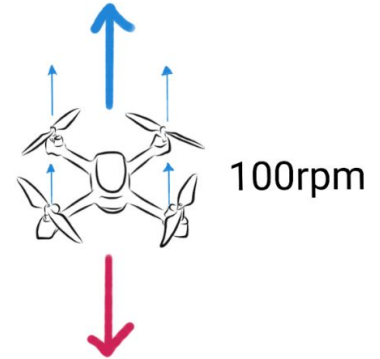


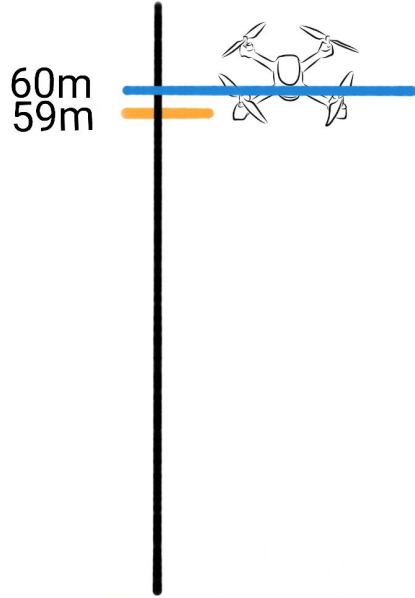
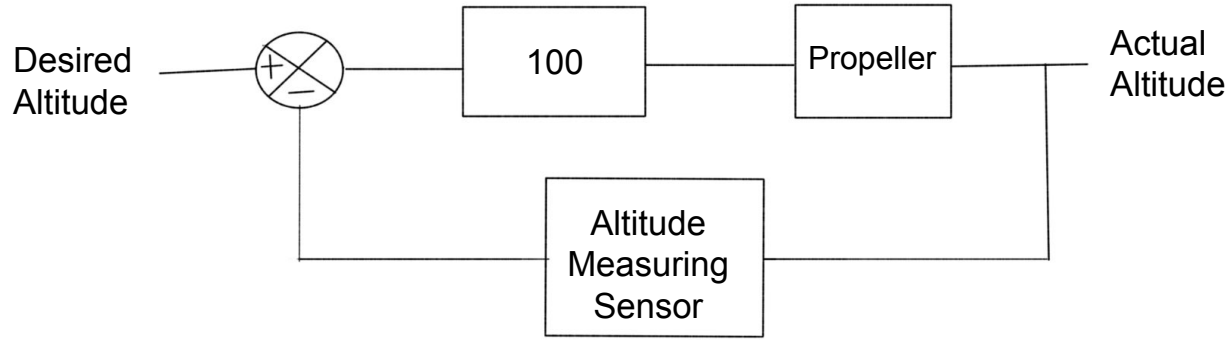




$$\begin{aligned}\text{Desired Altitude} - \text{Actual Altitude} \\ &= 60\text{m} - 60\text{m} \\ &= 0\end{aligned}$$

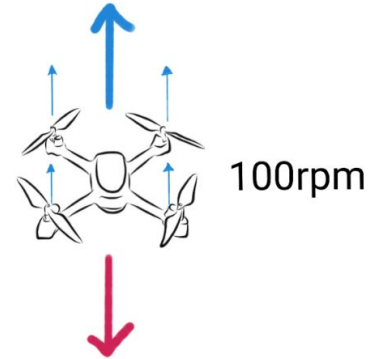
$$\text{Propeller Speed} = 0 * 10 = 0 \text{ rpm}$$

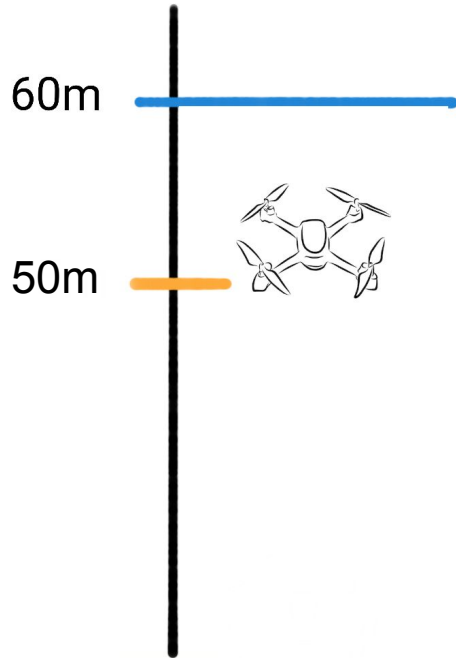
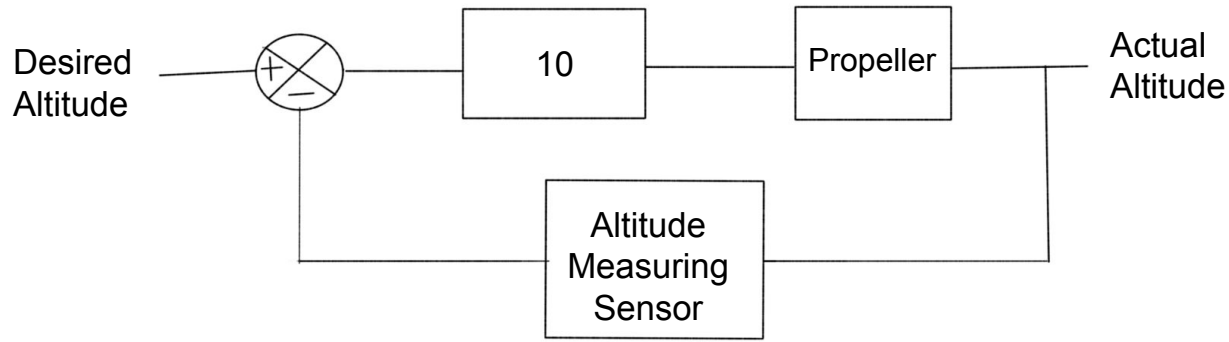




$$\begin{aligned}\text{Desired Altitude} - \text{Actual Altitude} \\ &= 60\text{m} - 59\text{m} \\ &= 1\end{aligned}$$

$$\text{Propeller Speed} = 1 * 100 = 100 \text{ rpm}$$

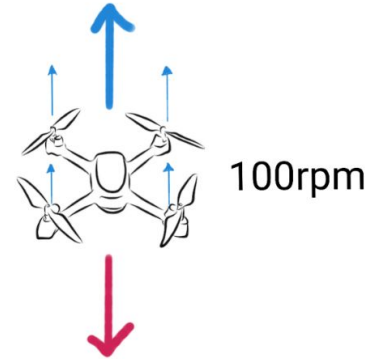


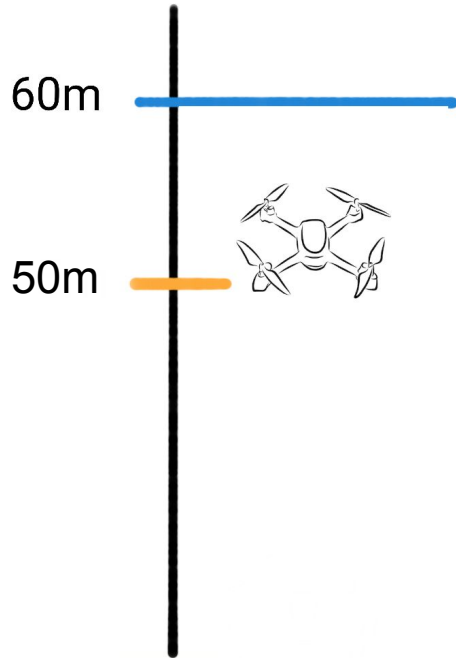
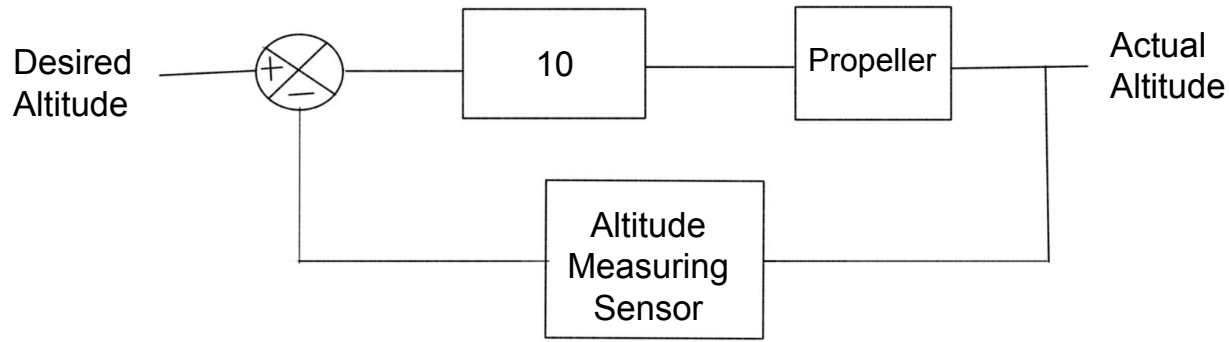


$$\text{Propeller Speed} = 10 * 10 = 100 \text{ rpm}$$

$$\text{Error} = 10$$

$$\text{Output} = K_p * \text{error}$$



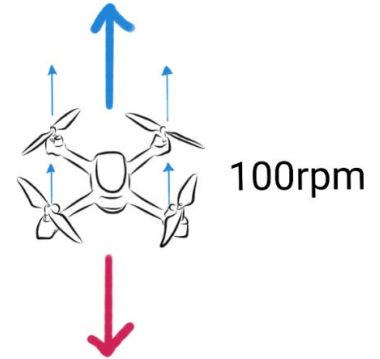


$$\text{Propeller Speed} = 10 * 10 = 100 \text{ rpm}$$

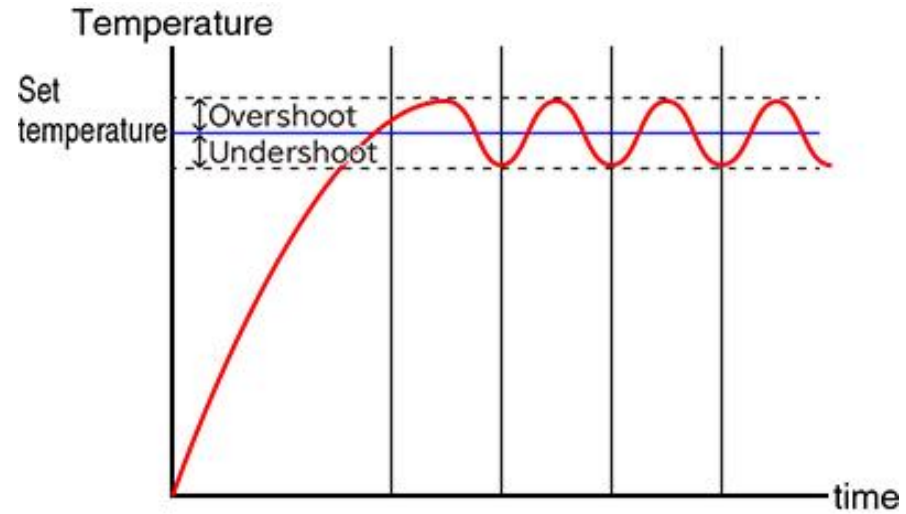
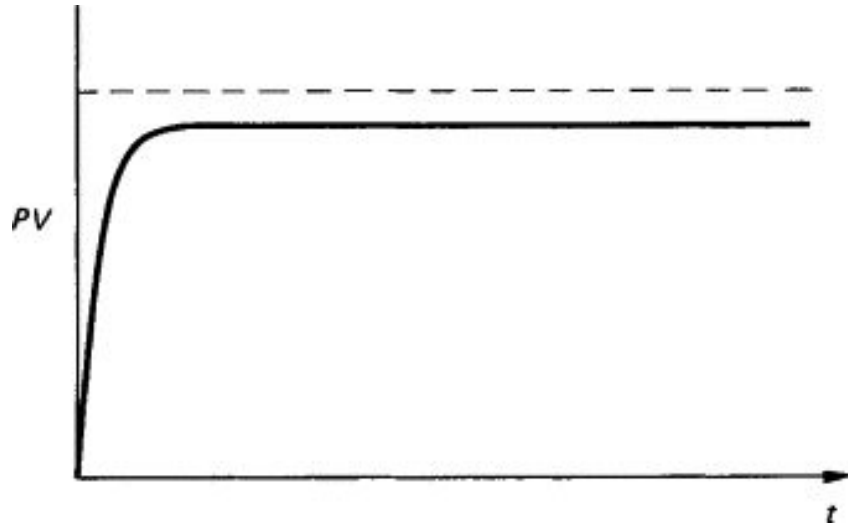
$$\text{Error} = 10$$

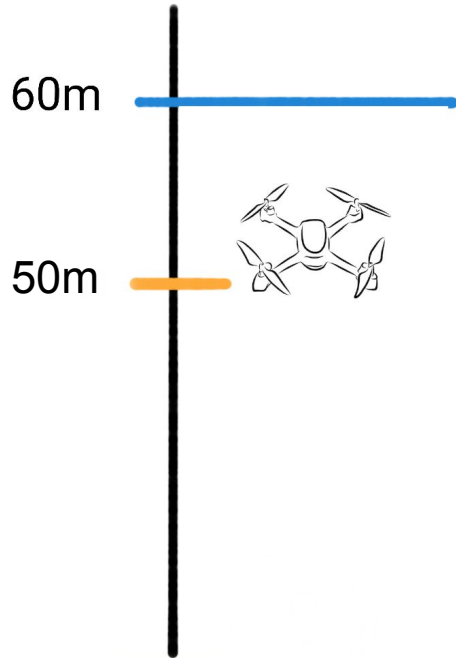
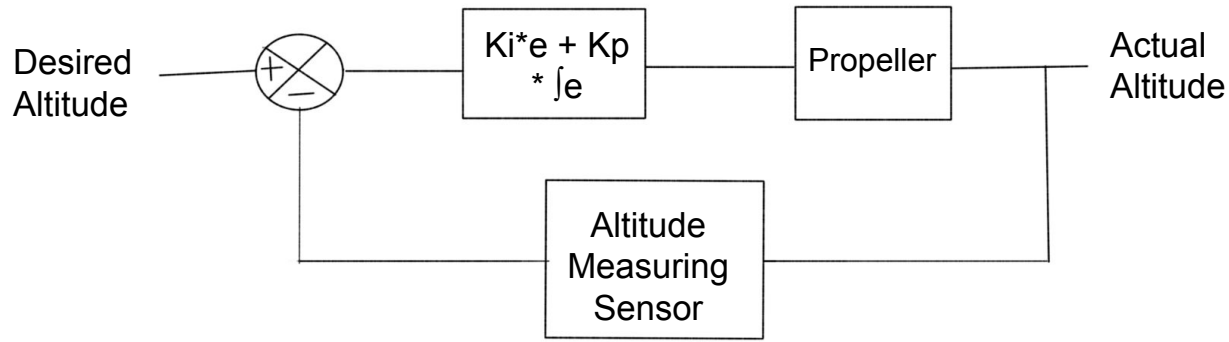
$$\text{Output} = K_p * \text{error}$$

Proportional Control



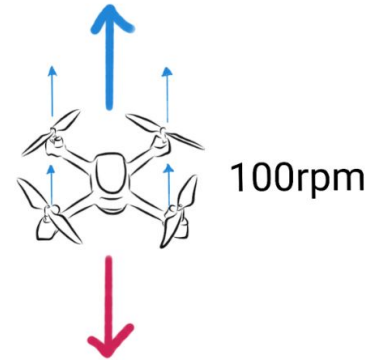
Proportional Control

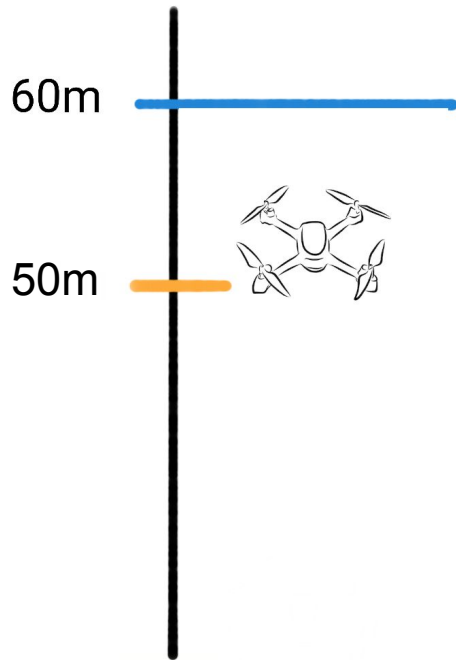
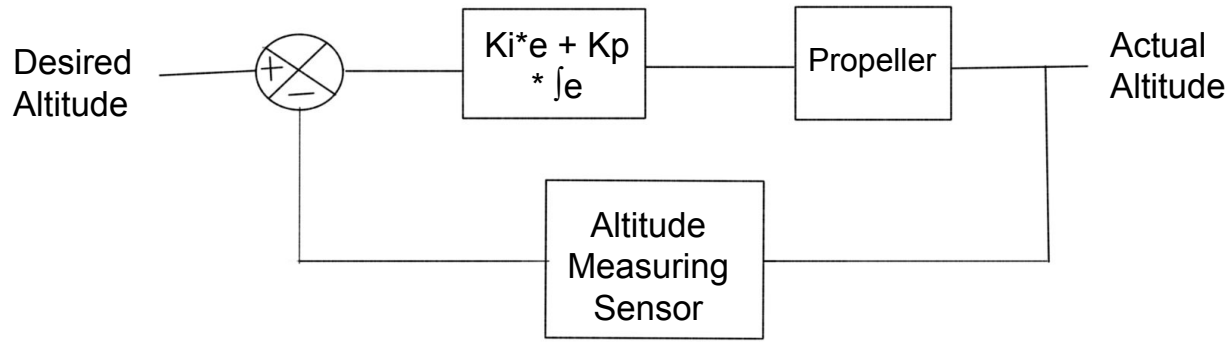




$$\text{Propeller Speed} = 10 * 10 = 100 \text{ rpm}$$

$$\text{Error} = 10$$

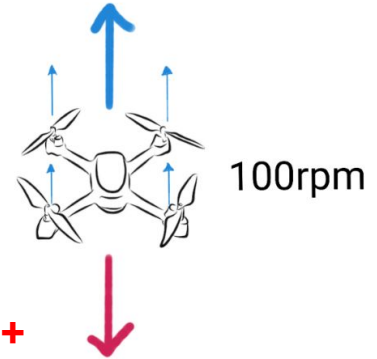


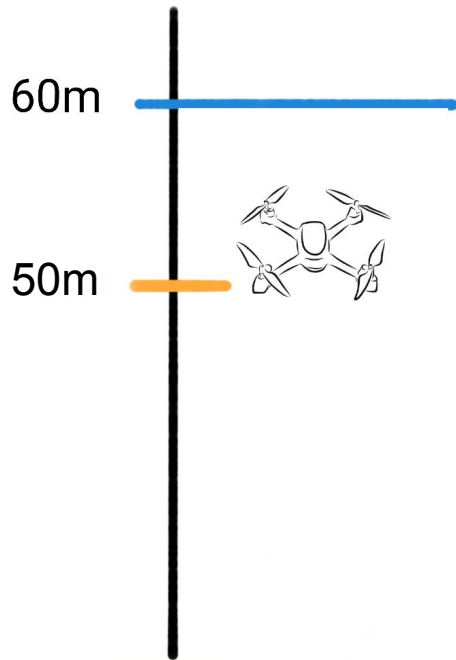
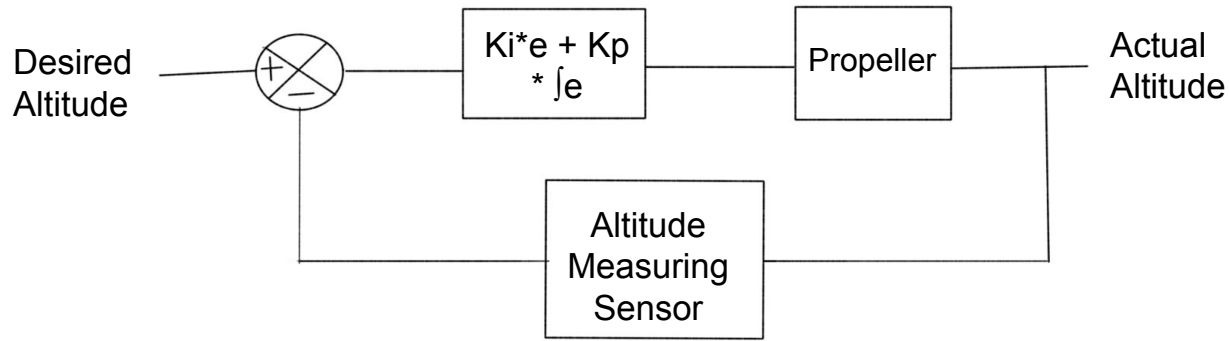


$$\text{Propeller Speed} = 10 * 10 = 100 \text{ rpm}$$

$$\text{Error} = 10$$

$$\text{Output} = K_p * \text{error} + K_i * (\text{error1} + \text{error2} + \text{error3})$$

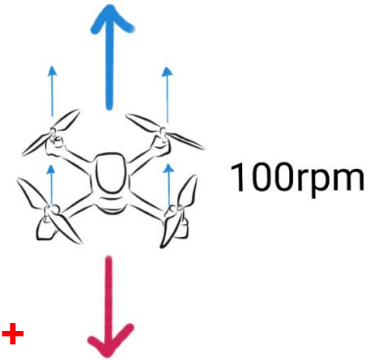




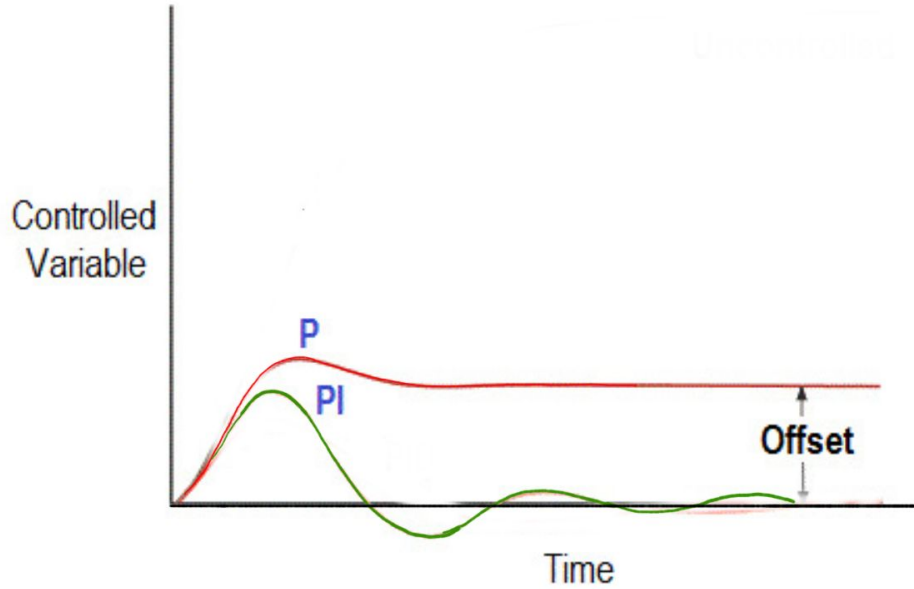
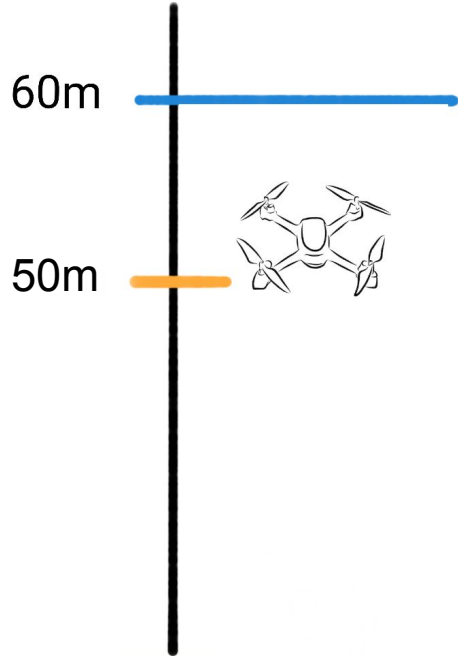
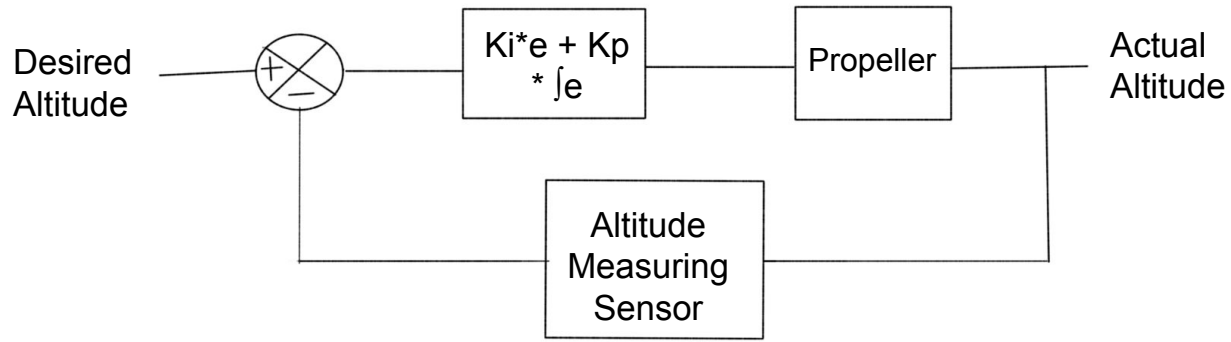
$$\text{Propeller Speed} = 10 * 10 = 100 \text{ rpm}$$

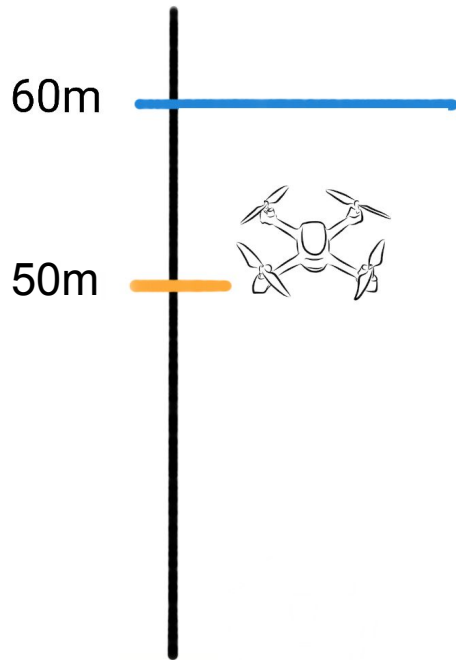
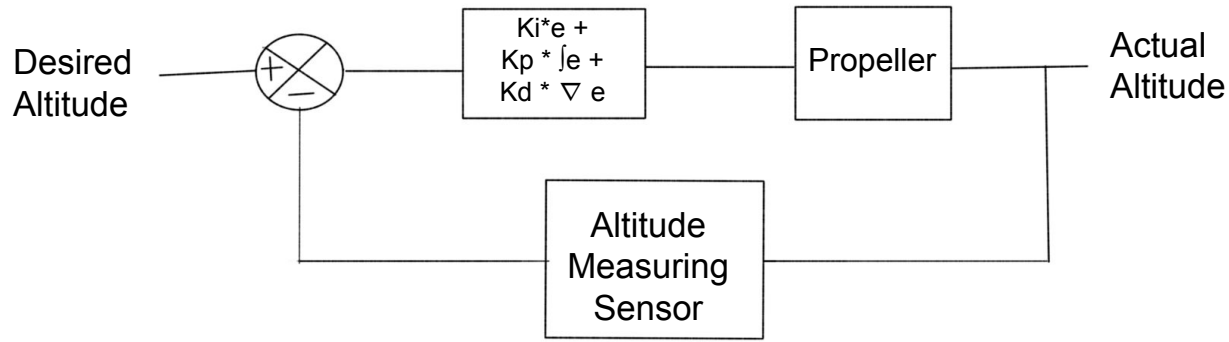
$$\text{Error} = 10$$

$$\text{Output} = K_p * \text{error} + K_i * (\text{error1} + \text{error2} + \text{error3})$$



Proportional - Integral Control

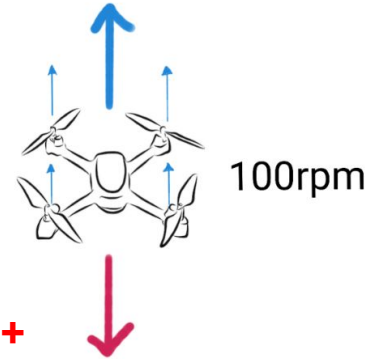


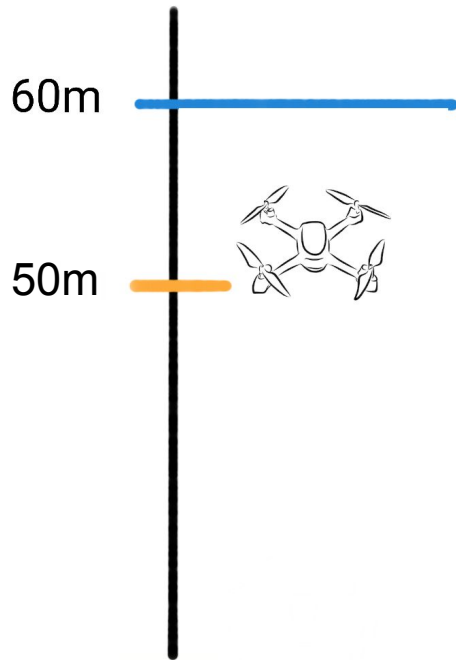
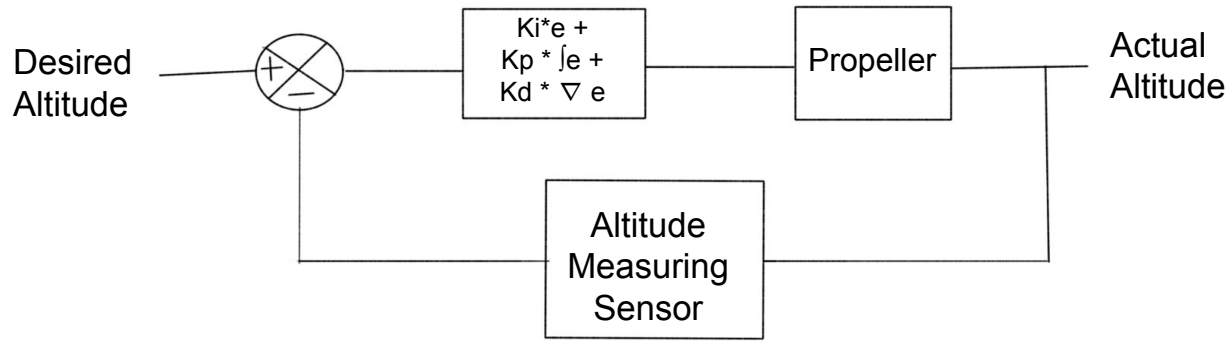


Propeller Speed = $10 * 10 = 100$ rpm

Error = 10

Output = $K_p * \text{error} + K_i * (\text{error1} + \text{error2} + \text{error3}) + K_d * (\text{error3} - \text{error2})$

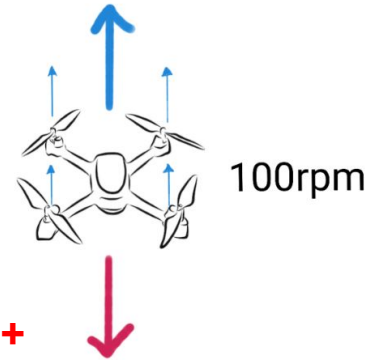




$$\text{Propeller Speed} = 10 * 10 = 100 \text{ rpm}$$

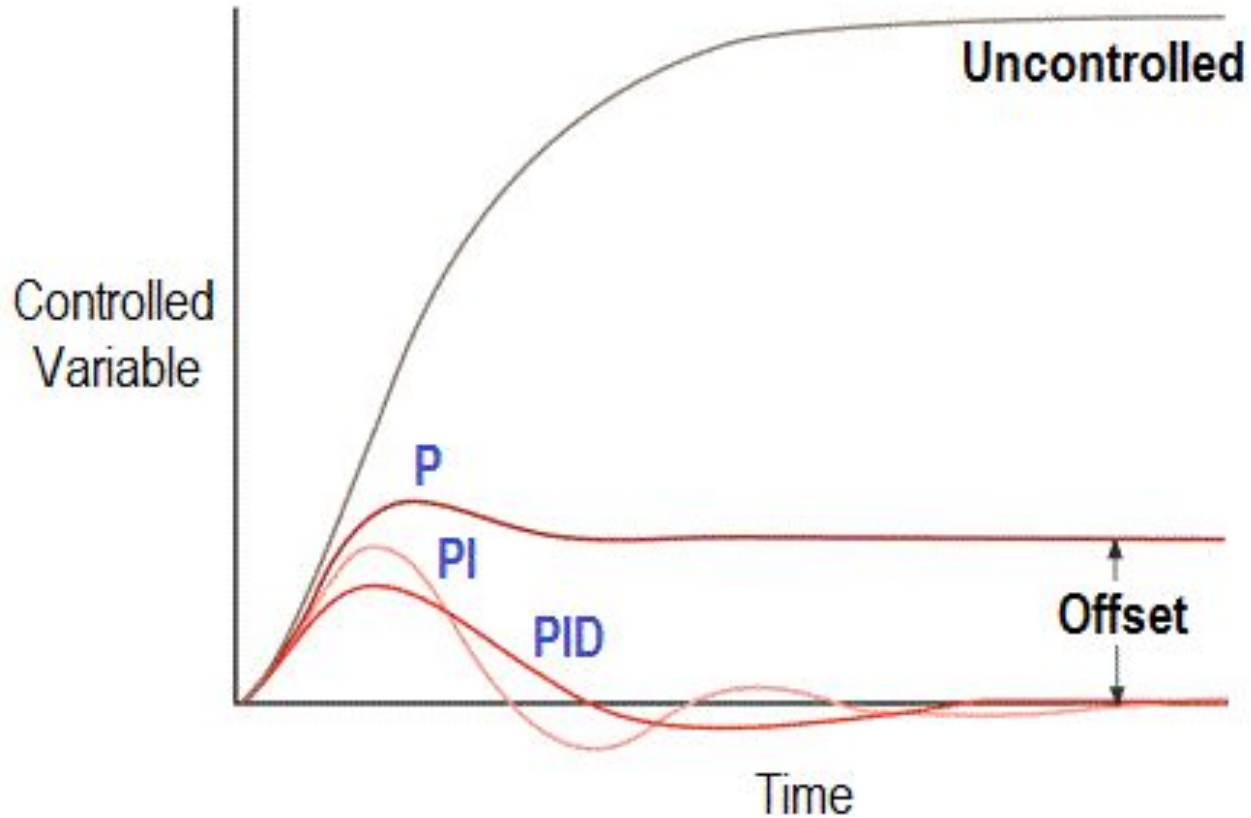
$$\text{Error} = 10$$

$$\text{Output} = K_p * \text{error} + K_i * (\text{error1} + \text{error2} + \text{error3}) + K_d * (\text{error3} - \text{error2})$$



Proportional - Derivative - Integral Control (PID) Control

Proportional - Derivative - Integral Control (PID) Control



PID Controller

Proportion al control : $u(t) = K_p e(t)$

Integral control : $u(t) = K_i \int_0^t e(t) dt$

Differenti al control : $u(t) = K_d \frac{d}{dt} e(t)$

- It produces an output, which is the combination of the outputs of proportional , integral & derivative controllers

$$u(t) \propto e(t) + \int e(t) + \frac{d}{dt} e(t)$$

$$\gg u(t) = K_P e(t) + K_I \int e(t) + K_D \frac{d}{dt} e(t)$$

How to get the PID parameter values ?

- If we know the transfer function, analytical methods can be used (e.g., root-locus method) to meet the transient and steady-state specs.
- When the system dynamics are not precisely known, we must resort to experimental approaches.

Ziegler-Nichols Rules for Tuning PID Controller

Using only Proportional control, turn up the gain until the system oscillates without dying down, i.e., is marginally stable. Assume that K and P are the resulting gain and oscillation period, respectively.

Then Use

for P control

$$K_p = 0.5 K$$

for PI control

$$K_p = 0.45 K$$
$$K_i = 1.2 / P$$

for PID control

$$K_p = 0.6 K$$
$$K_i = 2.0 / P$$
$$K_d = P / 8.0$$

Ziegler-Nichols Tuning
for second or higher
order systems

Generating PID parameters

- **Auto-Tuning Algorithms:**
 - Relay-based Auto-Tuning
 - A relay is introduced in the control loop.
 - Observes the system's oscillatory response.
 - Calculates PID parameters based on oscillation characteristics.
- **Trial and Error:** Iteratively adjust parameters based on system response.

Advantages and Applications of PID control

Advantages:

- Simple: easy to understand.
- Effective: Accurate and stable, even in dynamic environments.
- Robust: Can be adopted to different robot systems

Application:

- Arm positioning
- Robot Navigation
- Speed Control
- Balance control

Conclusion

- PID control---most widely used control strategy today
- Over 90% of control loops employ PID control, often the derivative gain set to zero (PI control)
- The three terms are intuitive---a non -specialist can grasp the essentials of the PID controller's action. It does not require the operator to be familiar with advanced math to use PID controllers
- Engineers prefer PID controls over untested solutions